1. A coupled partial melting Stokes-Darcy system. Strong form

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0\phi_f^{1+\Theta}}{\mu_f}\nabla(\phi_f^{-1/2}\tilde{q}_f) &= 0\\
\mu_s\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0\\
\nabla q - \nabla\cdot\hat{\sigma}(\mathbf{v}_s) &= -(1-\phi_f)\rho_r\mathbf{g}\\
\mu_s\nabla\cdot\mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0
\end{aligned}
\tag{1.1}
$$

List SI units

q $=$ pascal.

$\mathbf{v}$ $=$ m/s

Non dimensionalization choice

- $\check{\mathbf{v}} = \frac{\tilde{\mathbf{v}}_r}{k_0\rho_r g_y/\mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = (\frac{k_0\mu_s}{\mu_f})^{1/2}$

The resulting non dimensionalized form is therefore

$$
\begin{aligned}
\check{\mathbf{v}}_r + \phi_f^{1+\Theta}\nabla(\phi_f^{-1/2}\check{q}_f) &= 0\\
\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0\\
\nabla\check{q} - \nabla\cdot\hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f)\rho_r\hat{\mathbf{g}}\\
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0
\end{aligned}
\tag{1.2}
$$

scaled nondimensionalized weak form let $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla\cdot\mathbf{v}\mathrm{I}$

$$
\begin{aligned}
(2\phi_s\check{\sigma}(\mathbf{v}_{s,h}), \nabla\Psi_s) - (\check{q}_h, \nabla\cdot\Psi_s) &= -(\phi_s\check{\mathbf{f}}, \Psi_s)\\
(\nabla\cdot\check{\mathbf{v}}_{s,h}, \omega) + \Big(\frac{\hat{\phi}_f}{\phi_s}\check{q}_h, \omega\Big) - \Big(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_{f,h}, \omega\Big) &= 0\\
(\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r)) &= 0\\
(\hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \Big(\frac{1}{\phi_s}\check{q}_{f,h}, \omega_f\Big) - \Big(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h, \omega_f\Big) &= 0
\end{aligned}
\tag{1.3}
$$

1.1. Variational form and discretization.

1.1.1. Stokes and variational formulation.

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{1.4}$$

weak form

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{f}, \mathbf{v})_\Omega + < \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau >_{\Gamma_i} \\ &\quad + < \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu >_{\Gamma_i} - < \nabla \mathbf{u}_0, \nabla \mathbf{v} >_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{1.5}$$

let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have traction boundary condition term defined on Γ_i which is $< \mathbf{s}, \mathbf{v} >_{\Gamma_i}$. Traditionally a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0 \text{ on } \Gamma_u\}$ is employed here.

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + < \mathbf{s}, \mathbf{v}_h >_{\Gamma_{i,h}} - < \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h >_{\Gamma_{u,h}} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{1.6}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

1.2. Darcy.

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{1.7}$$

weak form

$$\begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - < p_0, v_n >_{\Gamma_i} - < \mathbf{u}_0, \mathbf{v} >_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{1.8}$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

1.2.1. A degenerate Darcy test problem.

$$\begin{aligned} \mathbf{v} + \phi \nabla (\hat{\phi}^{-1/2} p) &= f \\ \hat{\phi}^{-1/2} \nabla \cdot (\phi \mathbf{v}) &= g \end{aligned} \tag{1.9}$$

We manufacture a solution with degenerate porosity in the target physical domain $x, y \in [-1, 1]$.

$$\phi = \begin{cases} \cos^m(n(x^2 + y^2)) & x^2 + y^2 < \pi/2n \\ 0 & \text{otherwise} \end{cases} \tag{1.10}$$

where m is an even number, and $\pi/2n < 1$ another choice of test case is shown as follows:

$$\phi = \begin{cases} 0 & x \leq 0 \\ \phi_+ x^4 & x > 0 \end{cases} \tag{1.11}$$

Define an arbitrary velocity and pressure

$$\begin{aligned} \mathbf{v} &= [\frac{x}{5}, -y]^T \\ p &= \hat{\phi}^{1/2} \end{aligned} \tag{1.12}$$

The source term is calculated as follows:

$$f = \mathbf{v} \tag{1.13}$$

and the divergence term is:

$$\begin{aligned} g &= \phi^{-1/2}\phi_+ \nabla \cdot [\frac{x^5}{5}, -x^4 y]^T \\ &= \phi^{-1/2}\phi_+ (x^4 - x^4) = 0 \end{aligned} \tag{1.14}$$

a divergence free test case

1.2.2. Modified BR element. The error estimate

2. Linear system and Uzawa type algorithm. Boundary conditions are already included in the linear system formulation. Original linear system:

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix} \tag{2.1}$$

This is a double saddle point system. Symmetrically permutate original linear system and form a enlarged saddle point system.

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix} \tag{2.2}$$

(2.3)

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_d \end{bmatrix} \quad \tilde{\mathbf{C}} = \begin{bmatrix} \mathbf{C}_s & \mathbf{K} \\ \mathbf{K} & \mathbf{C}_d \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \end{bmatrix} \qquad \mathbf{G} = \begin{bmatrix} \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}$$

In this newly formed system, the $\tilde{\mathbf{A}}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration. Uzawa algorithm has been discussed extensively at [1], [2].

Consider the linear system of equation:

$$\begin{bmatrix} \mathbf{A} & -\mathbf{B} \\ -\mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ -\mathbf{G} \end{bmatrix} \tag{2.4}$$

Algorithm 2.1 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i - \mathbf{B}\mathbf{y}_i))$
3: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \mathbf{Q}^{-1}(\mathbf{G} - \mathbf{B}^T\mathbf{x}_{i+1} - \mathbf{C}\mathbf{y}_i)$

In the previous literature [2], $\mathbf{Q}$ is approximated with $\tau\mathbf{I}$, which works very well with the stokes part. However, it converges very slow with the same approximation when applied to Darcy part. Hence we proposed a more accuate Uzawa iteration procedure.

Algorithm 2.2 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i - \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{G} - \mathbf{B}^T\mathbf{x}_{i+1} - \mathbf{C}\mathbf{y}_i$
4: Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V} ||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

We use MINRES algorithm [?] due to schur complement being symmtetric and indefinite.

2.1. Convergence and stability analysis. Uzawa iteration is in fact a fixed point iteration procedure. Usually, uzawa iteration convergences slowly.

2.1.1. Anderson acceleration. Accelerating uzawa iteration has been discussed in the previously literatures. Anderson acceleration (Anderson mixing) [3]

3. Test cases. We test the linear system a little bit different from the linear system above.

3.1. Constant porosity and balanced pressure (divergence free velocity field).

3.2. Const porosity and unbalanced pressure.

$$\check{q} = 0 \qquad \check{q}_f = 2(x+y) \qquad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \qquad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \tag{3.1}$$

$$\nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y) \tag{3.2}$$

$$\mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \qquad (\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I} = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix} \tag{3.3}$$

$$\begin{aligned} \tilde{\mathbf{v}}_r + \phi_f^{1/2}\nabla \check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2}\begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2}\nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}\check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f}2(x+y) = 0 \\ \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}\check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\ -2(1-\phi_f)\nabla \cdot \sigma(\check{\mathbf{v}}_s) = -2(1-\phi_f)\nabla \cdot &(\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I}) = -\frac{8\phi_f^{1/2}}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned} \tag{3.4}$$

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